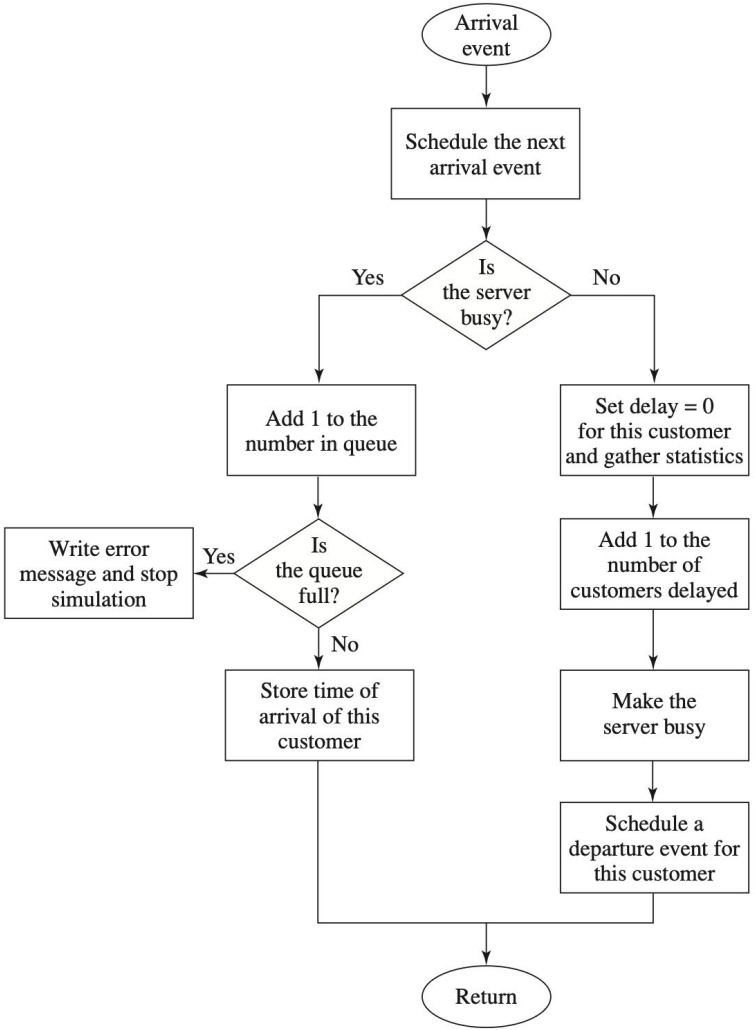
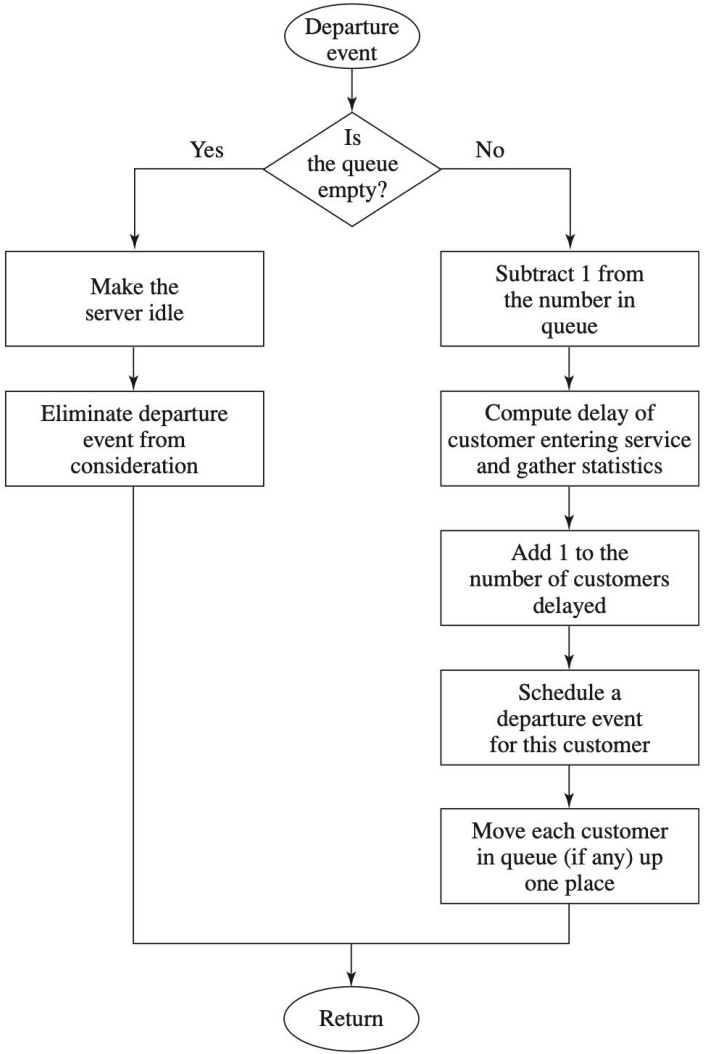
Simulación

Detalles de la simulación de una cola simple

Rutinas a considerar: Arribo



Rutinas a considerar: Partida



Salida de la simulación

```
Single-server queueing system

Mean interarrival time       1.000 minutes

Mean service time            0.500 minutes

Number of customers           1000


Average delay in queue       0.430 minutes

Average number in queue      0.418

Server utilization           0.460

Time simulation ended     1027.915 minutes
```

Inventarios

Seteo del problema

A company that sells a single product would like to decide how many items it should have in inventory for each of the next n months (n is a fixed input parameter). The times between demands are IID exponential random variables with a mean of 0.1 month. The sizes of the demands, D, are IID random variables (independent of when the demands occur), with

$$D = \begin{cases} 1 & \text{w.p. } \frac{1}{6} \\ 2 & \text{w.p. } \frac{1}{3} \\ 3 & \text{w.p. } \frac{1}{3} \\ 4 & \text{w.p. } \frac{1}{6} \end{cases}$$

La política (policy)

$$Z = \begin{cases} S - I & \text{if } I < s \\ 0 & \text{if } I \geq s \end{cases}$$

Los parámetros

K: costo de ordenar

h: costo de mantener

p: costo de pérdida

i: costo incremental por unidad

El nivel de inventario a lo largo del tiempo

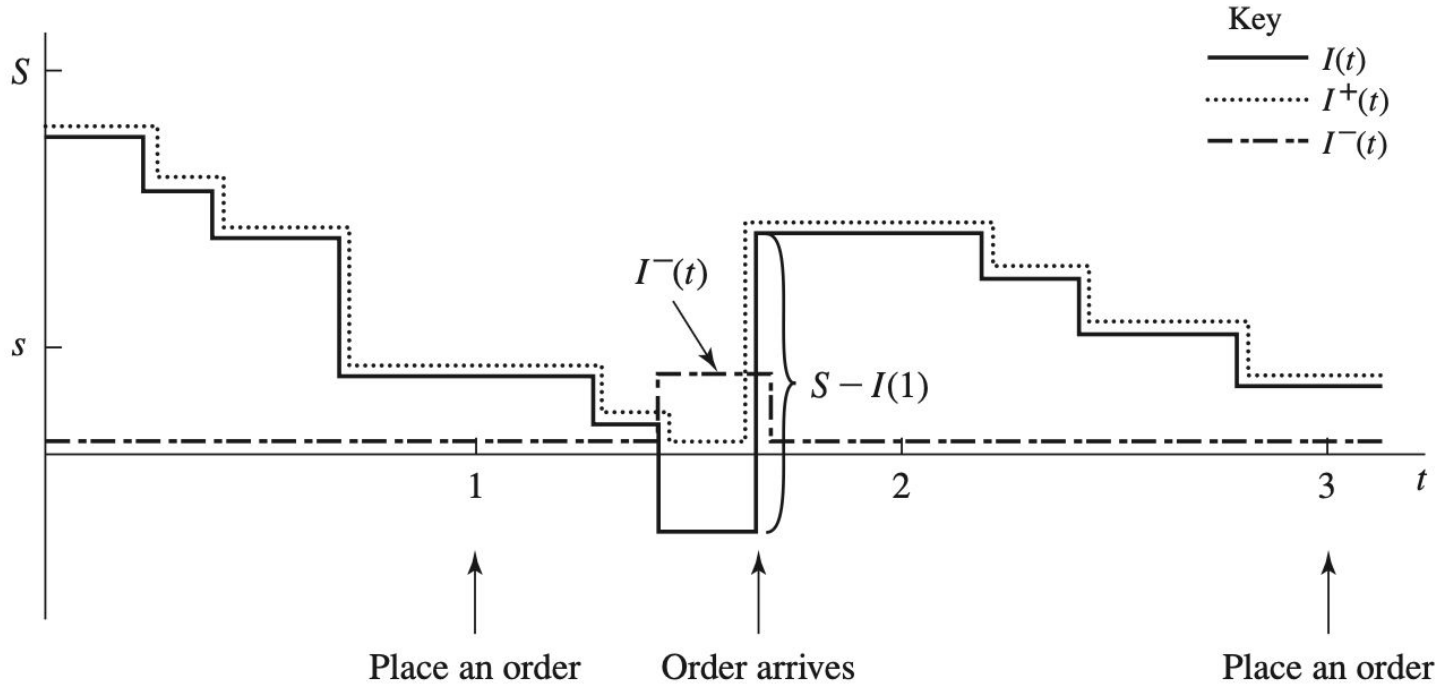
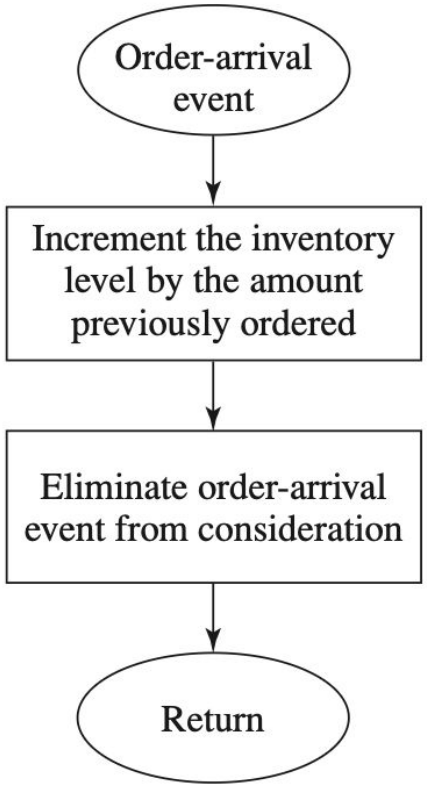
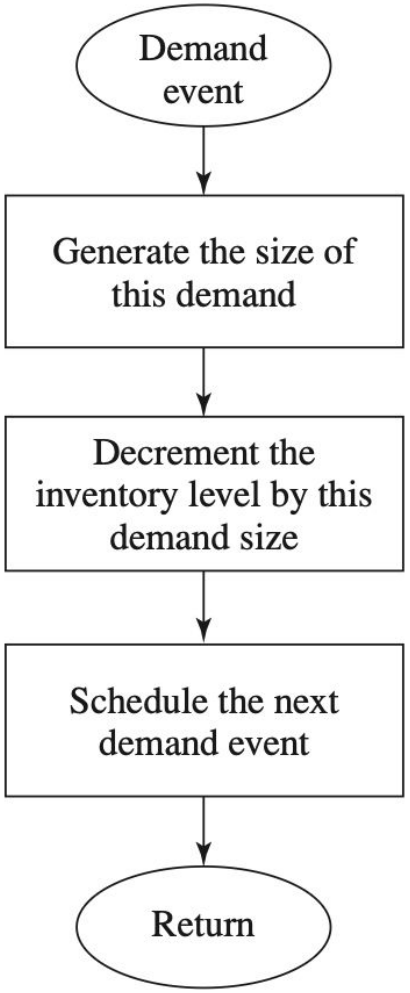


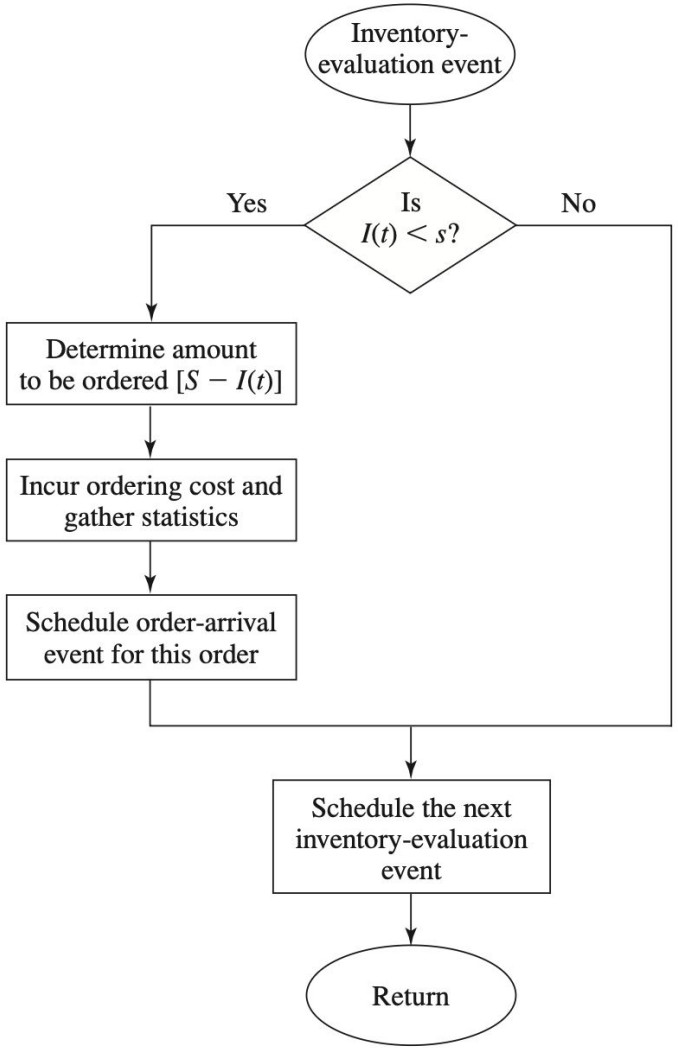
FIGURE 1.28
A realization of $I(t)$, $I^+(t)$, and $I^-(t)$ over time.

Los eventos:
Llegada de una orden
Demanda

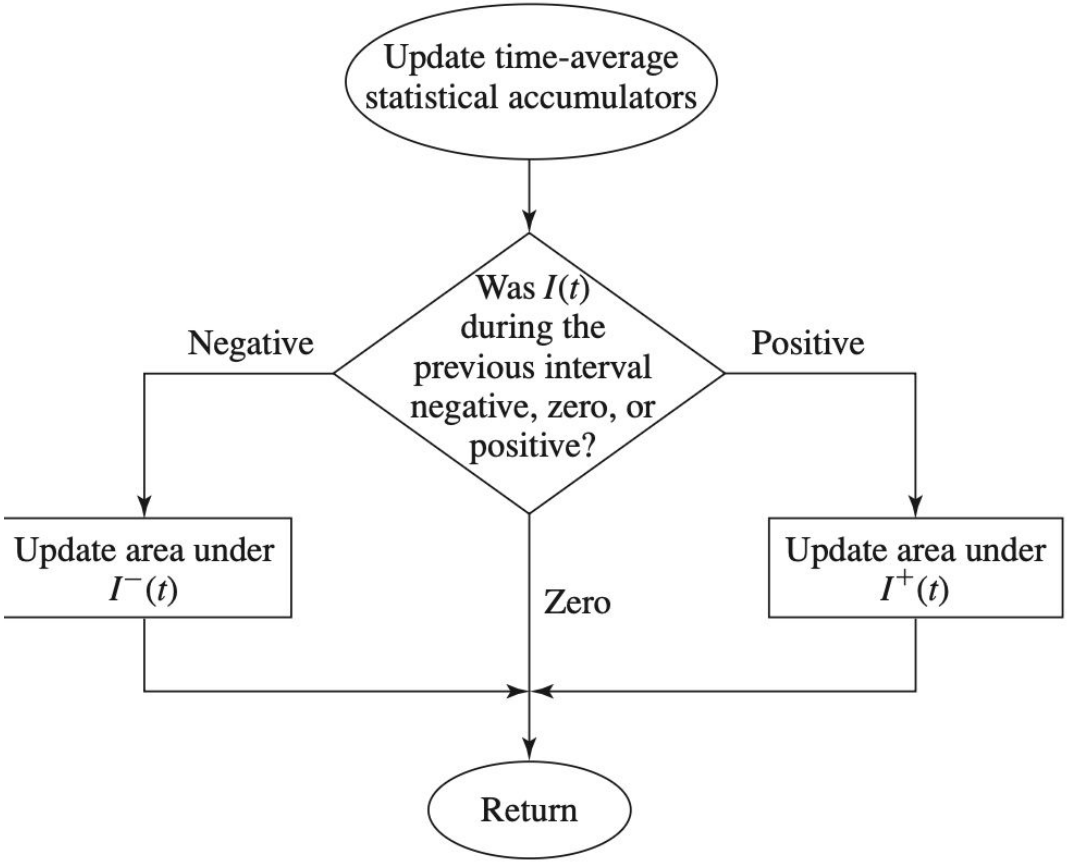




Los eventos:
Evaluación de inventario



Los eventos:
Actualizar contadores



Salida de la simulación

Single-product inventory system

Initial inventory level 60 items

Number of demand sizes 4

Distribution function of demand sizes 0.167 0.500 0.833 1.000

Mean interdemand time 0.10 months

Delivery lag range 0.50 to 1.00 months

Length of the simulation 120 months

K = 32.0 i = 3.0 h = 1.0 pi = 5.0

Number of policies 9

Policy	Average total cost	Average ordering cost	Average holding cost	Average shortage cost
(20, 40)	126.61	99.26	9.25	18.10
(20, 60)	122.74	90.52	17.39	14.83
(20, 80)	123.86	87.36	26.24	10.26
(20,100)	125.32	81.37	36.00	7.95
(40, 60)	126.37	98.43	25.99	1.95
(40, 80)	125.46	88.40	35.92	1.14
(40,100)	132.34	84.62	46.42	1.30
(60, 80)	150.02	105.69	44.02	0.31
(60,100)	143.20	89.05	53.91	0.24